

High Performance CMOS Static Logic Circuit Design

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Abstract

A high performance CMOS Static Logic implemented by a new circuit technique is presented. The gate outputs are preconditioned to minimize delay using a new clocking scheme and circuit design. The simulated CLA circuit shows that the average power-delay product is 1.62 times smaller than the static implementations for 0.25 μ m process.

1 Introduction

A static logic gate is one that has a well defined output once the inputs are stabilized and the switching transients have decayed away. Static CMOS logic gates are relatively easy to design and use. A static CMOS logic has many important properties: low sensitivity to noise and process variations, excellent speed, and low power consumption. With these important properties, static CMOS logic can be integrated with the CVTL technique to produce a high performance circuit.

In the following sections, a complex CMOS static gate with CVTL technique is described first. After the in-depth investigation of the design of static logic with CVTL technique, the larger integrated circuits are the primary design target. These circuits include adders, multipliers, and shifters. The speed of these elements often dominates the overall system performance. Hence, a careful design is needed. A Carry-Lookahead-Adder is, therefore, chosen to design with the CVTL technique to illustrate high performance ability, since the addition is probably the most commonly used arithmetic unit.

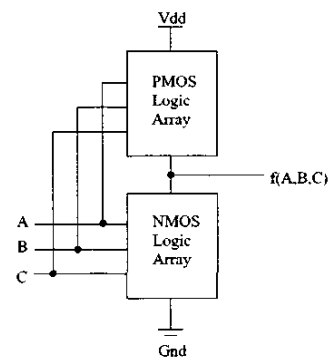


Figure 1: The general structure of a static CMOS logic gate

2 Complex CMOS static logic with CVTL technique

2.1 CMOS static logic design

A static complex CMOS gate is composed of two networks. One is pull-up network (PUN) and the other is pull-down network (PDN), which is shown in Fig. 1. The PUN and PDN networks should be designed so that one and only one of the networks is conducting in static steady state no matter what value of inputs.

2.2 Introduction of Critical Voltage Transition Logic (CVTL)

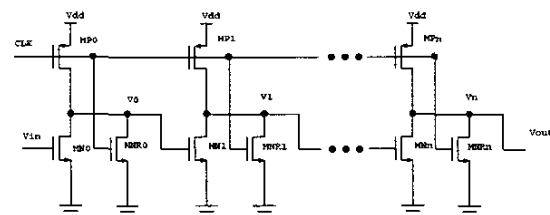


Figure 2: The basic CVTL configuration architecture

The CVTL technique is another circuit design style for CMOS logic gate. The principle of the CVTL technique is that the gate outputs are preconditioned to minimize delay using a new clocking scheme and circuit design. The Critical Voltage Transition Logic (CVTL) was first introduced in [1, 2]. The most important feature of CVTL is its unique speed improvement over other conventional logic families such as pseudo-NMOS, dynamic, complementary CMOS logic. In a conventional static CMOS logic inverter, the output of the logic gate does not begin a transition from Gnd to Vdd until the input reaches the 75 percent of Vdd . Therefore most of the transition time is wasted. The idea for improving the speed over other logic families is that CVTL logic precharges the output of the logic gate to a certain voltage, and then the output goes high or low according to input. Hence the time wasted in the conventional logic can be avoided. The basic CVTL inverter circuit configuration is illustrated in Fig. 2

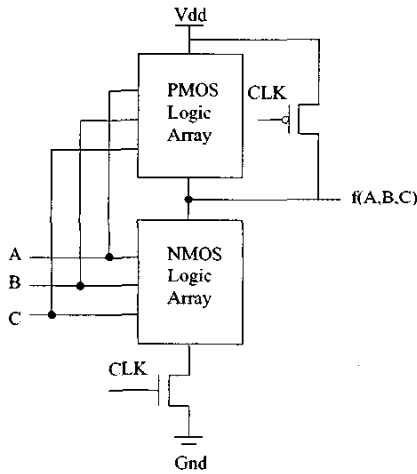


Figure 3: The general structure of a static CMOS logic gate with pull-down style CVTL technique

By applying the CVTL technique to static CMOS logic we provide another design direction which can preserve the performance advantage of both static and CVTL techniques. To integrate the CVTL technique into static CMOS logic, the clock controlled PMOS and NMOS of CVTL technique can be placed in a way such that there is no path between Vdd and Gnd . The example is a static CMOS logic gate with pull-down style CVTL technique. The clock controlled NMOS transistor is placed between the pull-down network and Gnd . The clock controlled PMOS is placed between the output node and Vdd . The static CVTL gate is pull-down style which is shown in Fig. 3. The output node is first precharged to Vdd when

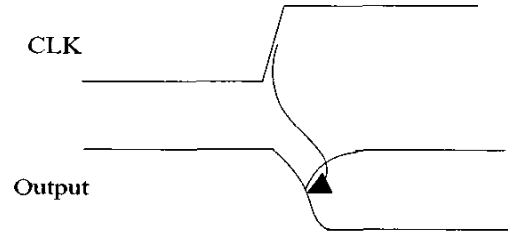


Figure 4: The timing diagram of static CMOS logic gate with pull-down style CVTL technique

the clock is low. Then the output is evaluated by both pull-up and pull-down network when the clock changes from low to high. The timing diagram of static logic with CVTL technique is shown in Fig. 4

3 16-bit Carry-Lookahead-Adder design

Integer addition is one of the simplest and most commonly used arithmetic operations and it is also the speed-limiting element. Therefore, careful optimization of the adder is a very important design issue in the basic arithmetic building block. This optimization can be done either at the logic level or circuit level. In the logic level the optimization is to rearrange the Boolean equation of adder so that a faster or smaller circuit can be achieved. Carry Lookahead Adder is an example of optimized logic in terms of speed. We will use Carry Lookahead Adder (CLA) as a design example.

A CLA adder requires one logic level to form p and g , two levels to form the carries, and two for the sum, for a total of five logic levels. This is an improvement over $2n$ levels required for the ripple-carry adder where n is the bit number of an adder. However, when the bit number n is larger, the CLA adder needs a fan-in of $n+1$ at the OR gates as well as the rightmost AND gate. Also the p_{n-1} signal must drive n AND gates. This makes it impractical to build a CLA adder when n is large. Therefore, we need to build an adder that has about $\log_2 n$ logic levels. We have seen that

$$c_1 = g_0 + c_0 p_0 \quad (1)$$

Similarly,

$$c_1 = G_{01} + P_{01} c_0 \quad (2)$$

G_{01} means that there is a carry-out generated out of the block consisting of the first two bits. P_{01} means that a carry propagates through this block. P and G have the following logic equations:

$$G_{01} = g_1 + p_1 g_0 \quad (3)$$

$$P_{01} = P_1 p_0 \quad (4)$$

More generally, for any j with $i < j, j+1 < k$, we have the recursive relations

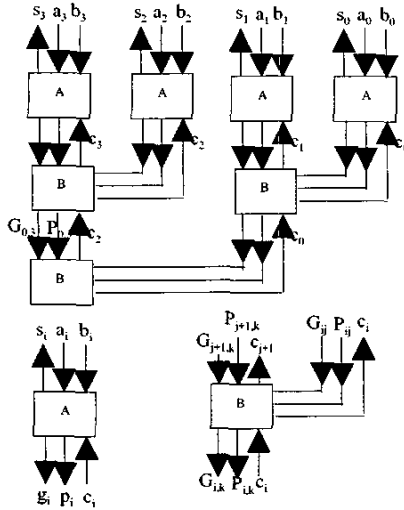


Figure 5: A complete CLA adder

$$c_{k+1} = G_{ik} + P_{ik} c_i \quad (5)$$

$$G_{ik} = G_{j+1,k} + P_{j+1,k} G_{ik} \quad (6)$$

$$P_{ik} = P_{ij} P_{j+1,k} \quad (7)$$

The complete CLA adder implemented by these equations is shown in Fig. 5

The numbers to be added flow into the top and downward through the tree, combining with c_0 at the bottom flowing back up the tree to form the carries. Note that a small piece of extra logic to compute c_4 for the carry-out of the adder is missed. Fig. 5 is a 4-bit adder where block A and B are the circuits to generate the corresponding p and g . The inputs and outputs of these two blocks are shown in the bottom of Fig. 5. The relationship of inputs and outputs is in Equ. 5, 6, and 7. This 4-bit CLA adder can easily extend to 8-bit, 16-bit, or 32-bit adder. For example, an 8-bit adder can be formed by two 4-bit adder with an additional block B.

The logic design of the CLA adder is described in a previous subsection. The critical path of a CLA adder is the one to compute the carry-out signal. This can be easily verified from Fig. 5. The static logic with CVTL technique can be implemented into this CLA adder. First,

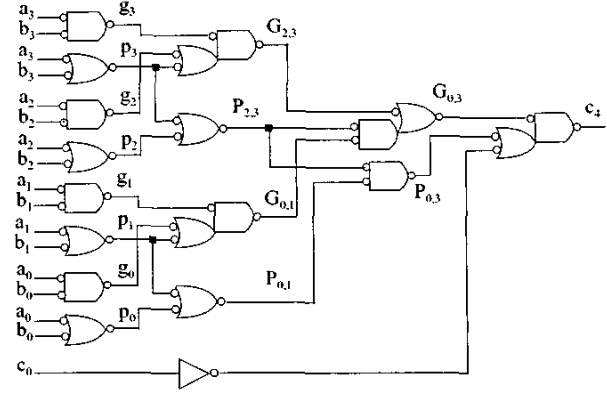


Figure 6: A 4-bit CLA adder gate level diagram

the gate level diagram of Fig. 5 is shown in Fig. 6. Block A is the first level of p and g . Block B is the second and third level of P and G . The carry-out is the fourth level. For 8-bit and 16-bit CLA adders, the critical path is 5 and 6 logic levels. The circuit design of a CLA adder using static logic with CVTL technique is in Figs. 7, and ??.

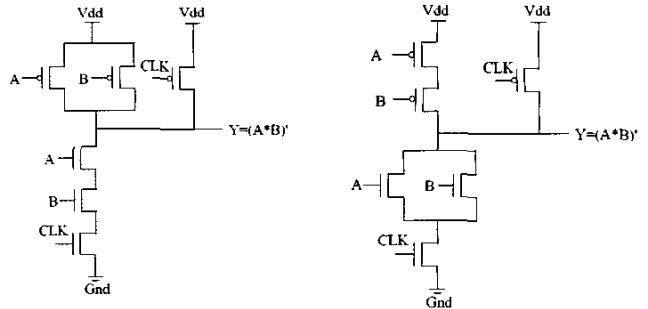


Figure 7: CVTL CLA adder circuit level design: NAND gate(left) and NOR gate(right)

In [3], multi-level CVTL NOR circuits have distortion from the third level and the following levels. This is also true in the static logic with CVTL technique. To resolve the distortion caused in each logic gate, the delay clock scheme is employed. The delayed clock element is inserted to each level of logic to insure that each output of CLA can be in the stable state. The major impact after employing the clock delay scheme is the speed improvement of the circuit. In the delayed clock implementation,

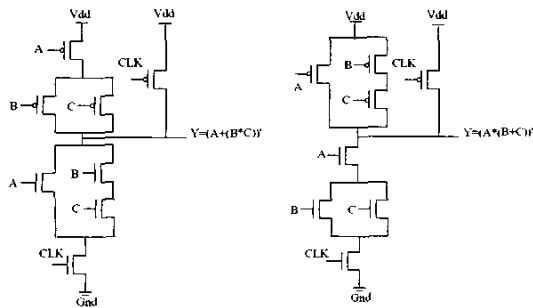


Figure 8: CVTL CLA adder circuit level design: AOI gate(left) and OAI gate(right)

each level of CLA is evaluated by a delayed clock from the clock of the previous level.

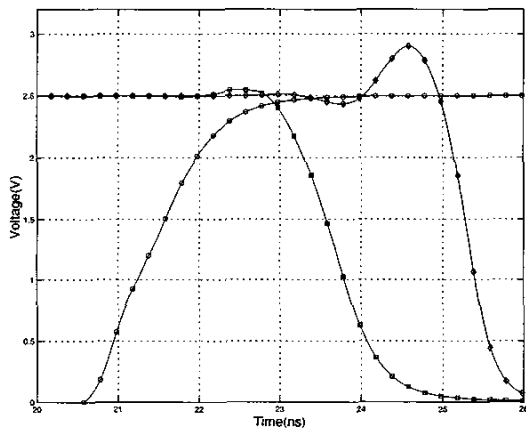


Figure 9: Speed simulation comparison: CMOS static logic with pull-down style CVTL technique ($\square$), CMOS static logic ($\diamond$)

The propagation delay of CLK to output is 0.2242ns in the static logic with CVTL technique and 0.4094ns for the static logic which is the delay time from input to output. The speed-up of static logic with CVTL over static logic is 1.826.

The power consumption is 28.31mw in the static logic with CVTL technique and 25.17mw for the static logic. The power consumption of static logic with CVTL is 12.50 percent more than static logic. The power delay products of static logic with CVTL technique and static logic are 6.347pj and 10.3046pj respectively. This implies that power delay product of static logic with CVTL technique is 1.62 better than the power delay product of static logic.

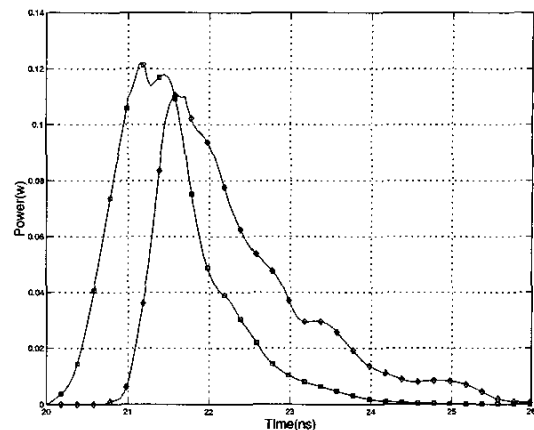


Figure 10: Power consumption comparison: CMOS static logic with pull-down style CVTL technique ($\square$), CMOS static logic ($\diamond$)

4 Conclusion

A static logic with CVTL technique is investigated in this paper. CVTL is a new logic style in which outputs are precharged to between V_{dd} and G_{nd} with a new clock scheme and circuit design. The simulation result of simple CMOS complex gate shows that CVTL technique can still be valid in the static logic style. The technology file used for simulation is $0.25\mu m$ TSMC process. A CLA adder is also implemented with static logic with CVTL technique. The results show that the new circuit can have 1.8 times speedup over static logic counterpart with 12 percent more power overhead. The power overhead is primarily contributed by the clock delay element. The overall power delay product of static logic with CVTL technique is still better than static logic.

References

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